



VMRS Chatbot Jan 2026

REYES FAMILY
OF BUSINESSES



Today's Goal

Introduce how we used,

- Knowledge Graph (Neo4j) to structure domain context for AI agents
- Claude Code + LLM-as-a-Judge to build and test AI Agents

with the VMRS use case as a case study.

Context

- Fleet Master Data Management Problem: Vendors assign different part numbers and descriptions to functionally equivalent parts (e.g., 1 part, 4 vendors, 4 different part numbers). This requires operator labor to identify equivalent parts.
- Current Workflow: Fleet team manually searches internal systems + vendor websites for each part with ~30 min/part and dependence on 2-3 domain experts. Operators look for the industry standard code for each part known as the VMRS (Vehicle Maintenance Reporting Standards) code.
- Use Case Goal: Reduce part identification time from ~30 minutes to ~5 minutes (83%) using an AI knowledge graph chatbot.

PART	DESCRIPTION	MANF_PARTMFR_NAME	VMRS
49KD86	BLUE ARRPW LIGHT	ECCO	034-004-253
01-4934-81	LAMP, MARKER, LED, AMBER	GROTE INDUSTRIES, INC.	034-004-253
GG78285	LAMP, AMBER REFLECTIVE SEALED	GRAND GENERAL ACCESSORIES MANUFACTURING	034-004-253
A06-40578-000	LIGHT, CLEARANCE LED - M2	FREIGHTLINER CORPORATION	034-004-253

Example: Lighting System, Lamp assembly, and LED - Marker Lamp component

Why choose a knowledge graph (KG)?

- Traditional SQL / ML method:
 - Vendors use different names/descriptions for each part -> no standard to match against
 - No way to include operator knowledge as context
- Knowledge Graph + LLM:
 - Maps different vendor descriptions to one standard VMRS code
 - Reasoning over structured relationships in graph
 - Human-in-the-Loop: Human operators provide context ("The part is under the front left of the hood") to narrow options
 - (Planned) Improved performance over time: confirmed matches add synonyms and context

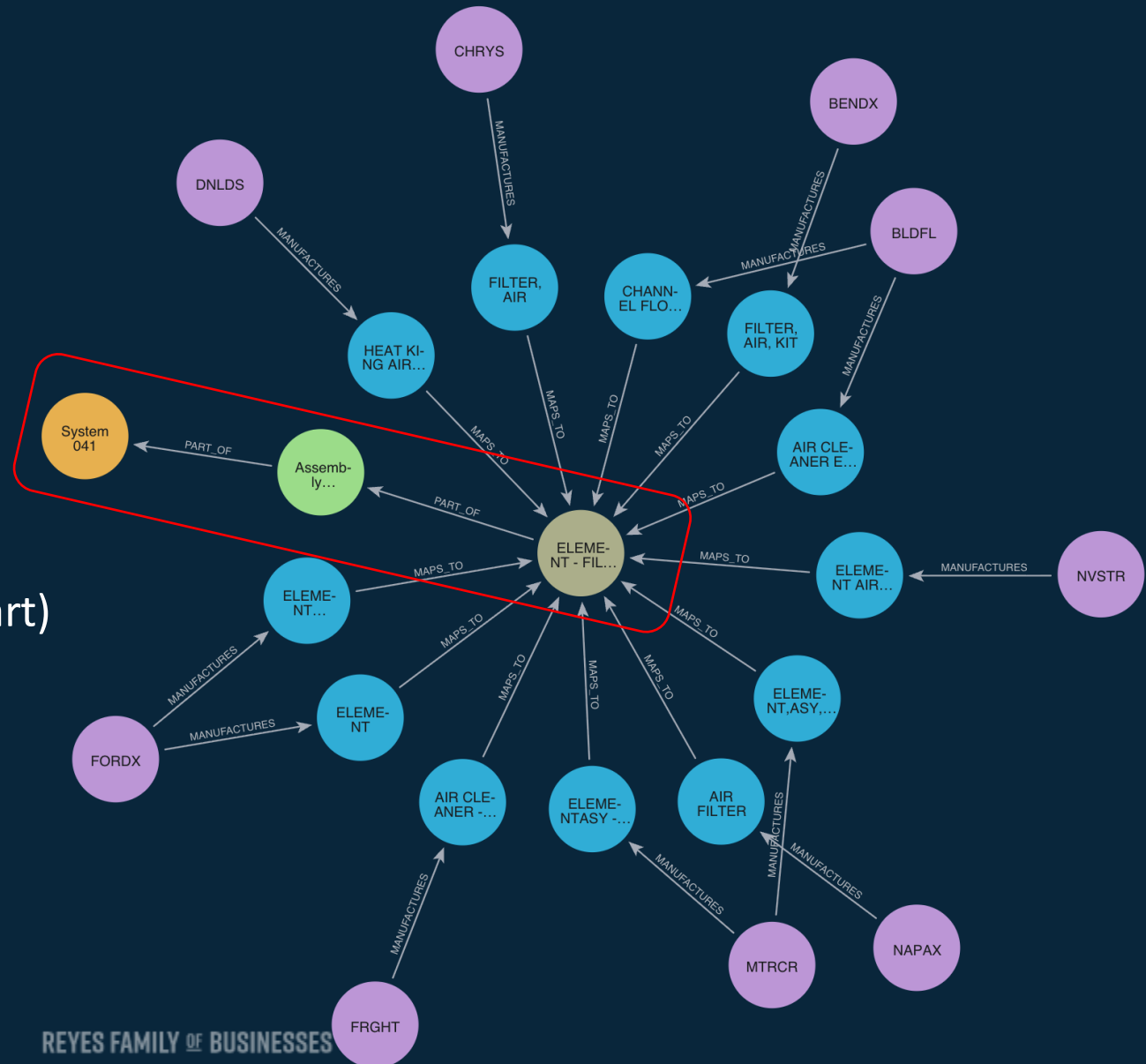
VMRS Codes in the KG: 1 VMRS Code to Many Part relationship

Entities:

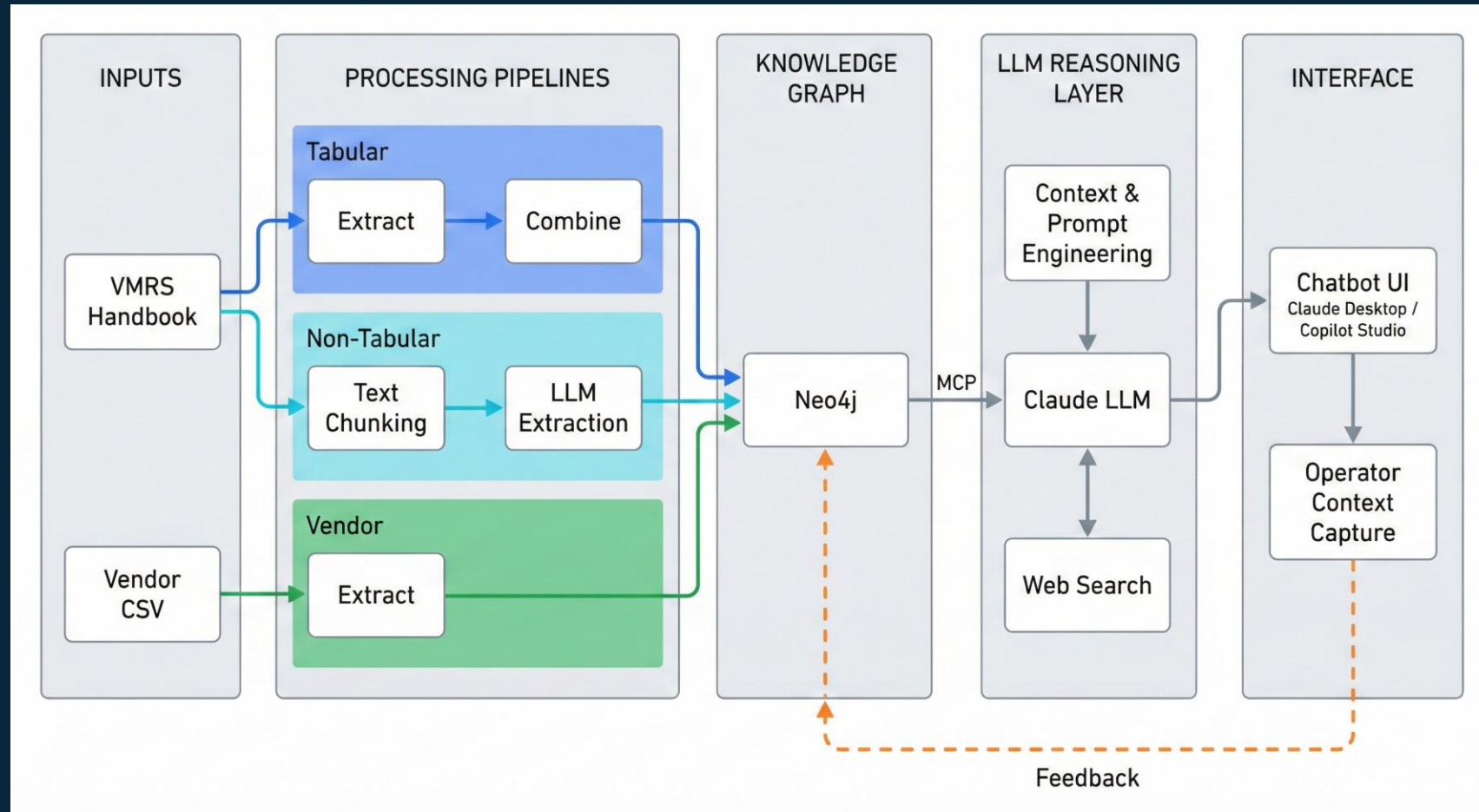
- 1 VMRS code (system, assembly, component)
- 12 parts (Blue)
- 9 Vendors (Purple)

Relationships:

- (part)-[PART_OF]-(part)-[PART_OF]-(part)
- (part)- [MAPS_TO]-(VMRS code)
- (vendor)-[MANUFACTURES]-(part)



Knowledge Graph Chatbot Architecture



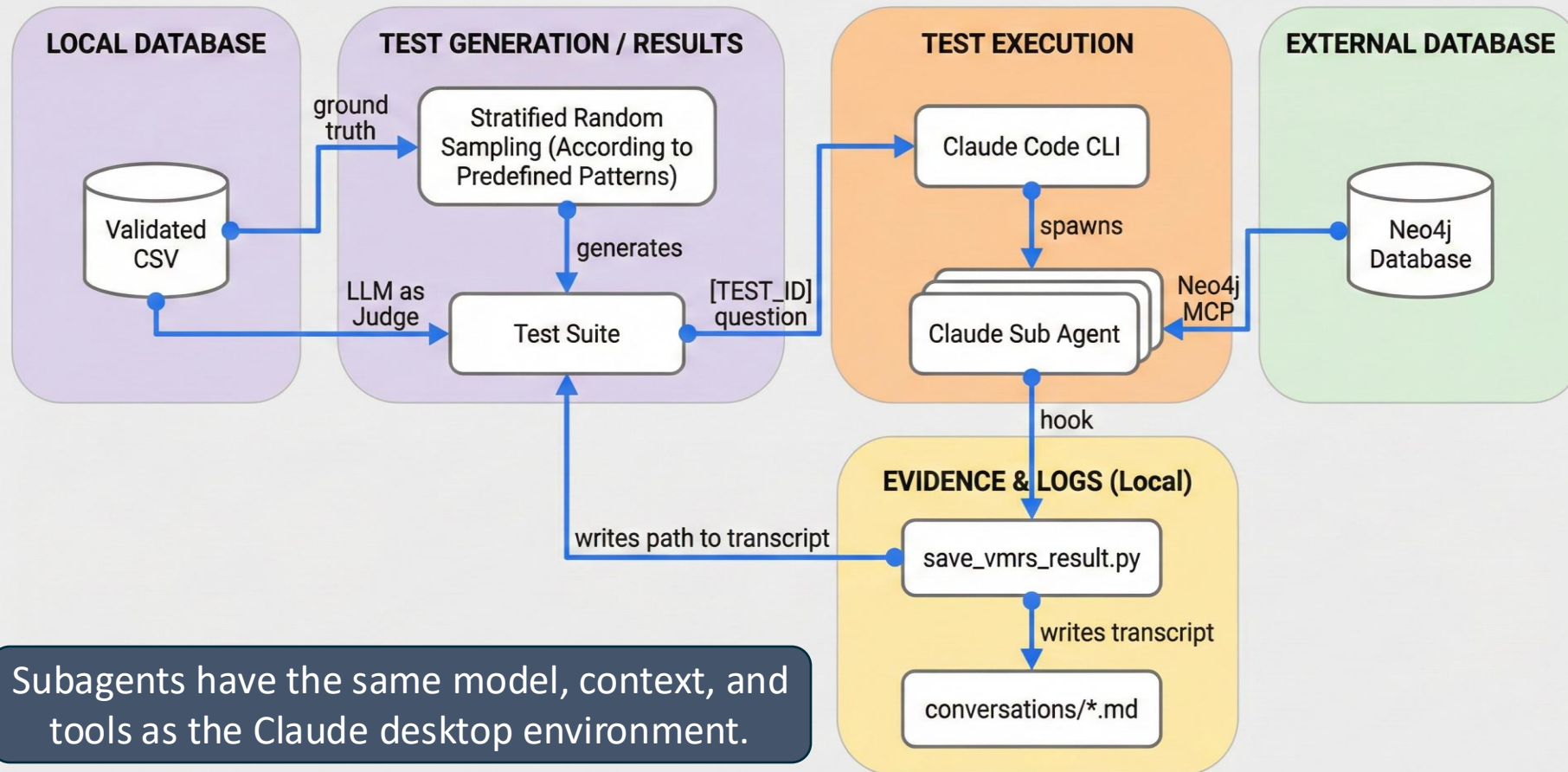
Demo

- Pattern 1: Search by description
 - What is the VMRS code for grille?
 - What is the VMRS code for flux capacitor?
- Pattern 2: Search by part number
 - What is the VMRS code for part number 2234788PE?
- Pattern 3: Search by part number + Vendor
 - What is the VMRS code for WALTCO LIFT CORP part WO-10099360?
- Pattern 4: Verify if 2 parts are the same
 - Do parts HG0067 and S-32097 have the same VMRS code?

Testing

- 73 / 80 (91%) test cases pass with 9 following patterns tested
- 68 / 70 (96%) "look up" test cases pass
- 2 look up fails
 - Description Partial Look up: DRIVERS SIDE SEAT (partial) vs DRIVERS SIDE SEAT BELT (full)
 - Test case bug: Vendor name selected from wrong column for test case
- 5 Validation – invalid Fails
 - Codes exist in VMRS hierarchy but don't map to any parts: LLM reasoning layer change needed

Testing Architecture



Subagents have the same model, context, and tools as the Claude desktop environment.

Next Steps

- Phase 1: Complete PoC
 - Developed test suite
 - Finetune web search feature
 - Develop HitL feature
- Phase 2: Prototype
 - Develop prototype in Reyes tech-stack

FAQ

- What was the biggest bottlenecks in this project?
 - Finding the correct data model
 - Getting the correct VMRS codes
- What patterns were tested?
 - Next slide
- Business impact?
 - Refer to the savings / business impact

Appendix

Tested Patterns

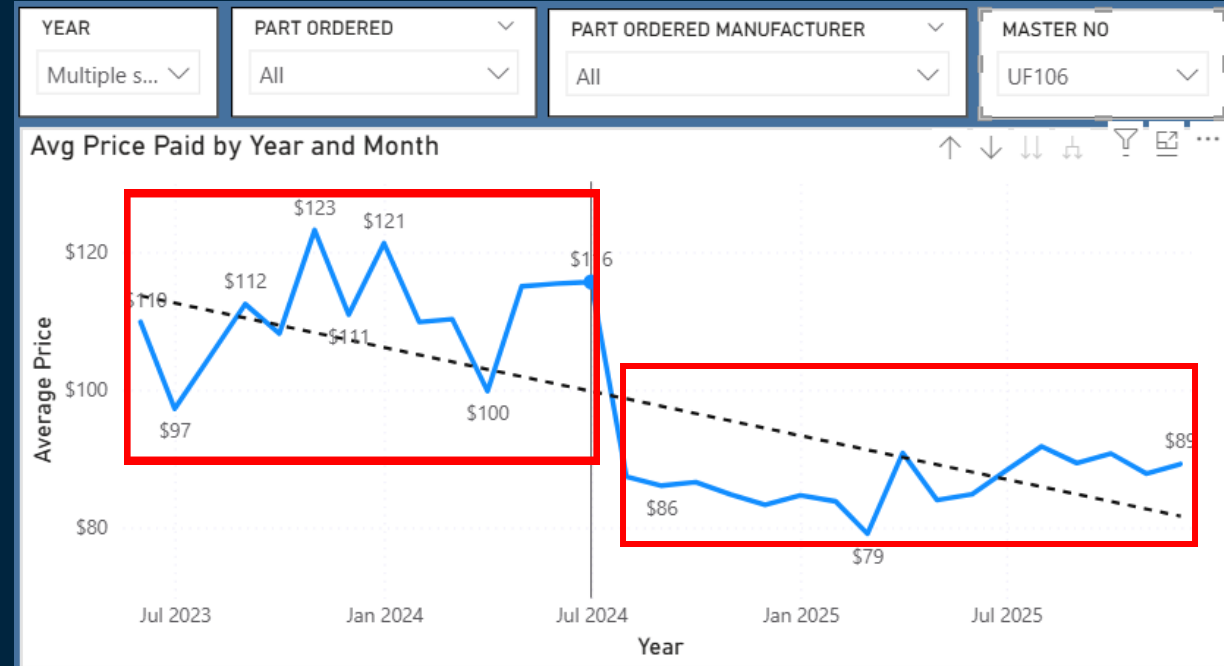
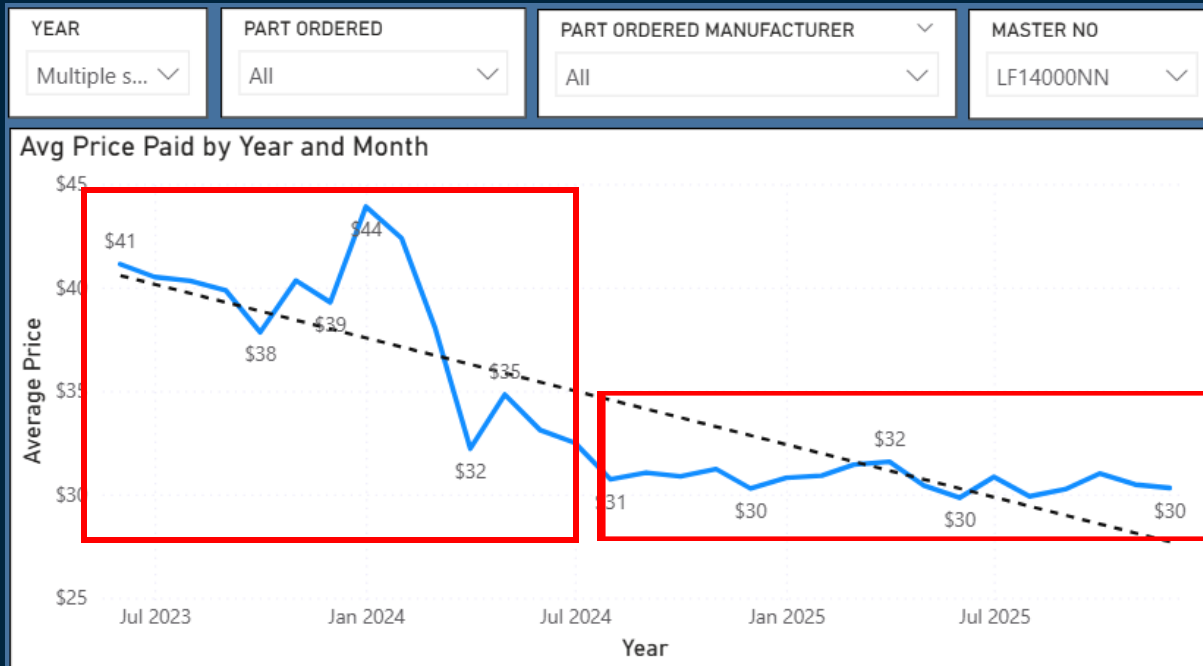
- Look up VMRS code given part number
- Look up VMRS code given vendor/manufacturer + part number
- Look up VMRS code given exact part description text
- Look up VMRS code given partial part description text
- Look up VMRS hierarchy path (System - Assembly - Component) given a VMRS code
- Look up vendor parts given a VMRS code
- Confirm whether a VMRS code is valid given a known-valid VMRS code
- Confirm whether a VMRS code is valid given a “known-absent”
- Look up whether two parts share the same VMRS code given two part numbers

What is the business impact?

- Future labor savings (\$62.5k):
 - o Annual labor savings for part identification: \$62.5k
 - Assumptions: 5k yearly parts, \$30/hr, 30 minutes -> 5 minutes
- Previous filter part savings (\$160k):
 - o Previous vendor optimization of ~240 parts saved \$160k annually.
 - o 0.8% (out of 30,000) coverage saved \$160k

Filter examples: LF14000NN: \$8 (20%) average saved per part

UF106: \$40 (32%) saved per part



Savings Details

- Purchase Optimization: One previous optimization over 240 parts allowed \$160k in annual savings.
- Data cleanup cost: 1 time fee of ~\$20,000 for outsourced VMRS code cleanup to 3rd party company and is not 100% accurate, with an additional \$500 to \$1,250 quarterly costs.
- Up to \$18.75k quarterly Labor Cost: Quarterly, 1,250 parts require 15-30 minutes each to research/validate, labor hrs. 312-625, labor cost and ~\$9,400 - ~\$18,750

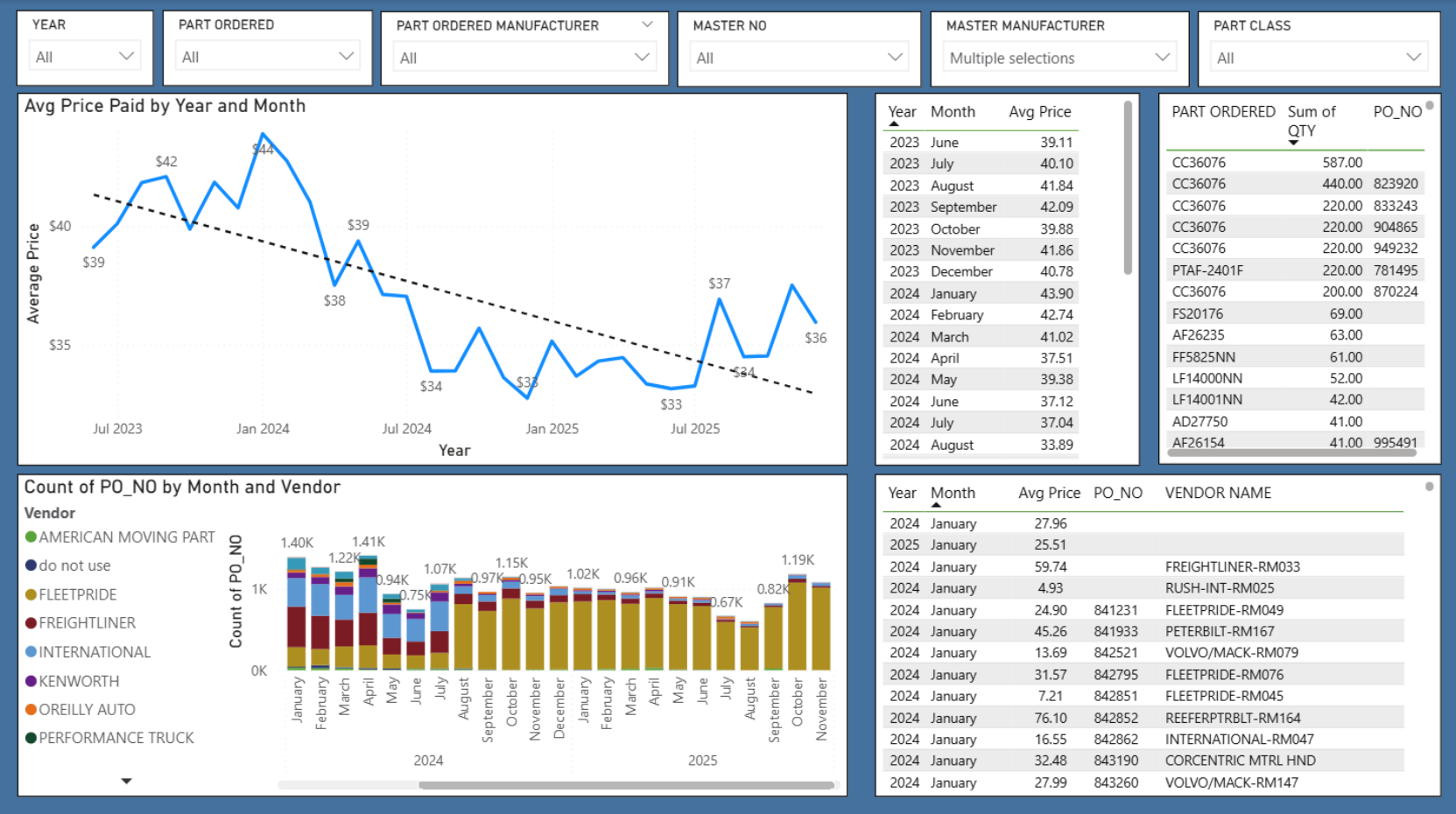
Link to handbook

- Link to handbook: [VMRS Industry knowledge](#)

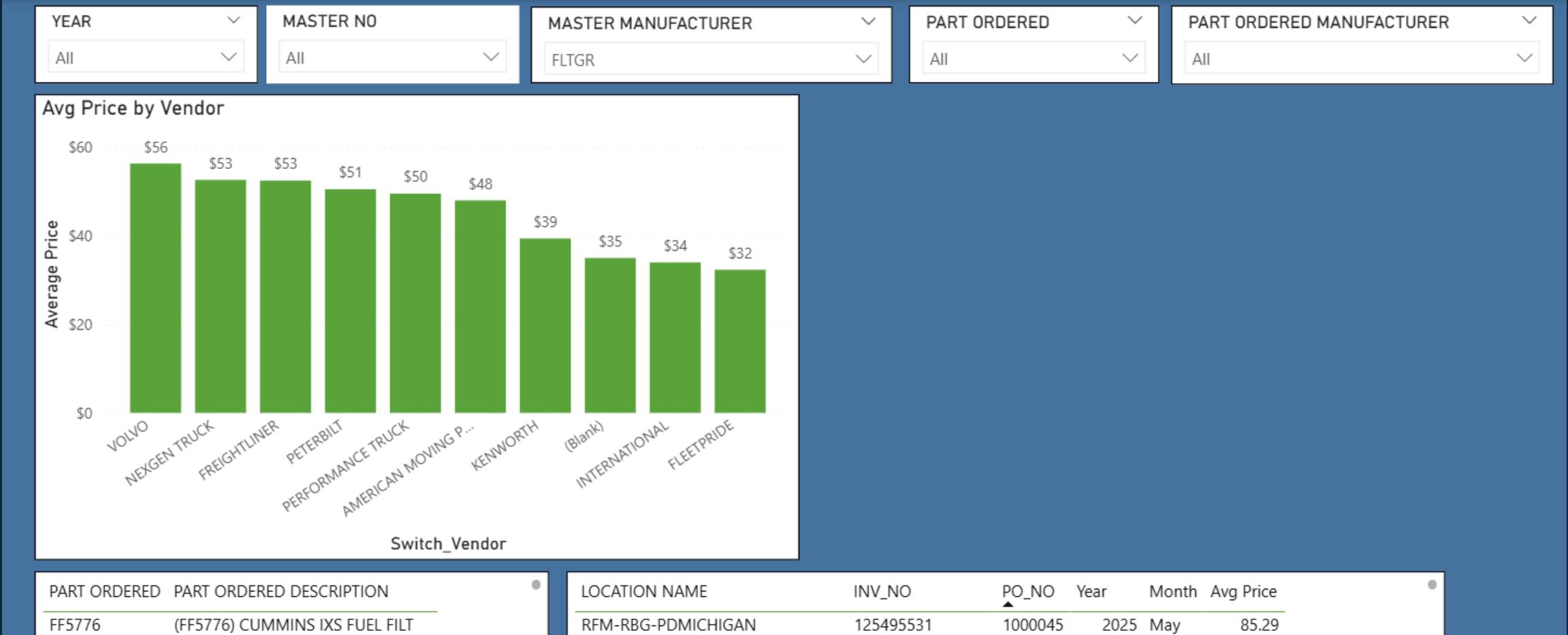
Savings and Cost

- Previous Initiative
 - Alternative Filter Vendors Annual Savings: ~\$160,000
- Current Labor Annual Cost: \$75,000
 - 5,000 parts * .5 hours per part * \$30 per hour
- Parallel Costs
 - Data Cleanup One Time Fee: ~\$20,000
 - Data Cleanup Annual Cost: \$5,000
 - \$1,250 quarterly
 - Expanded coverage: 240 vs 19,000 parts

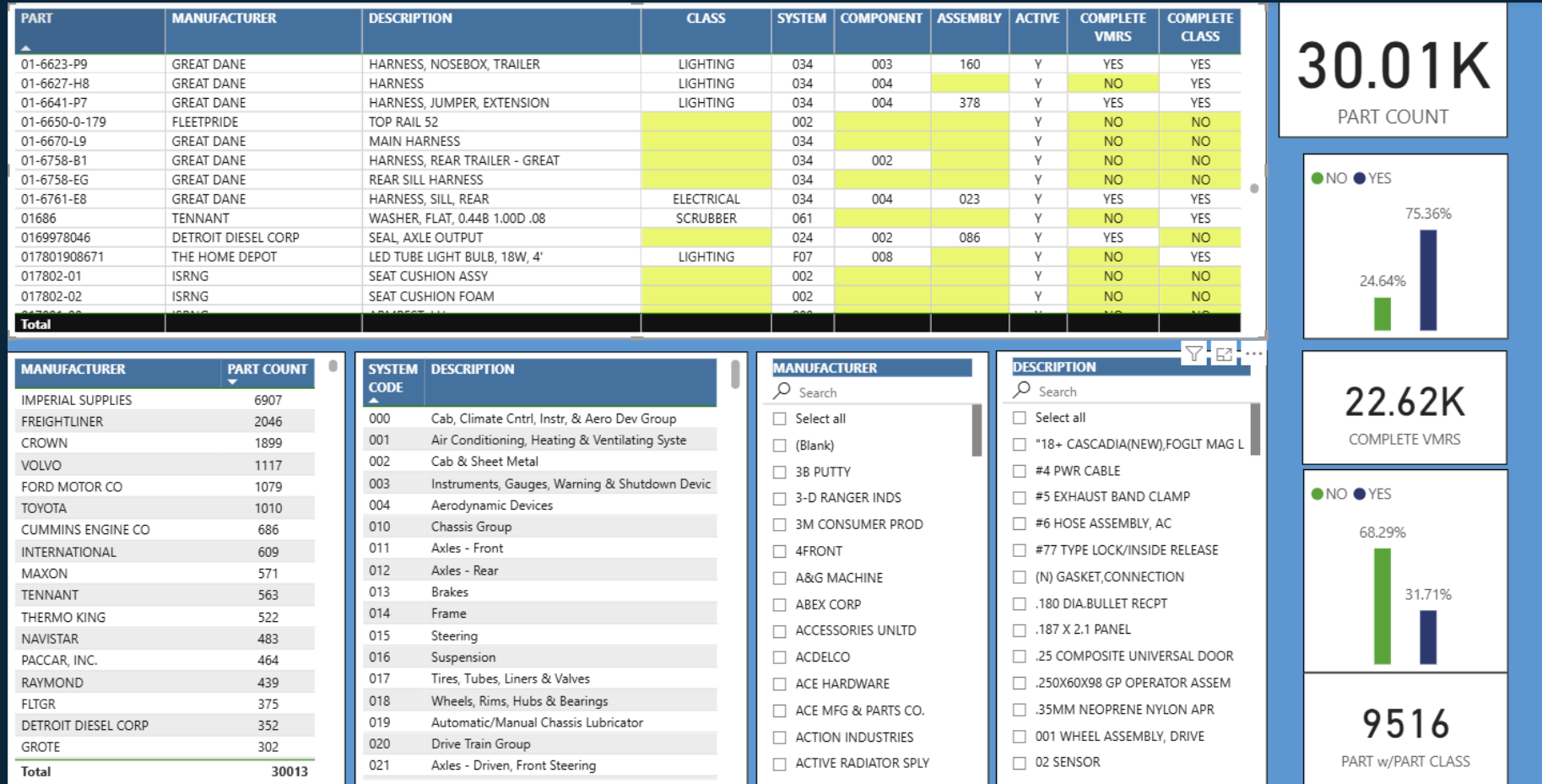
Previous Initiative



Previous Initiative



What does the data look like?



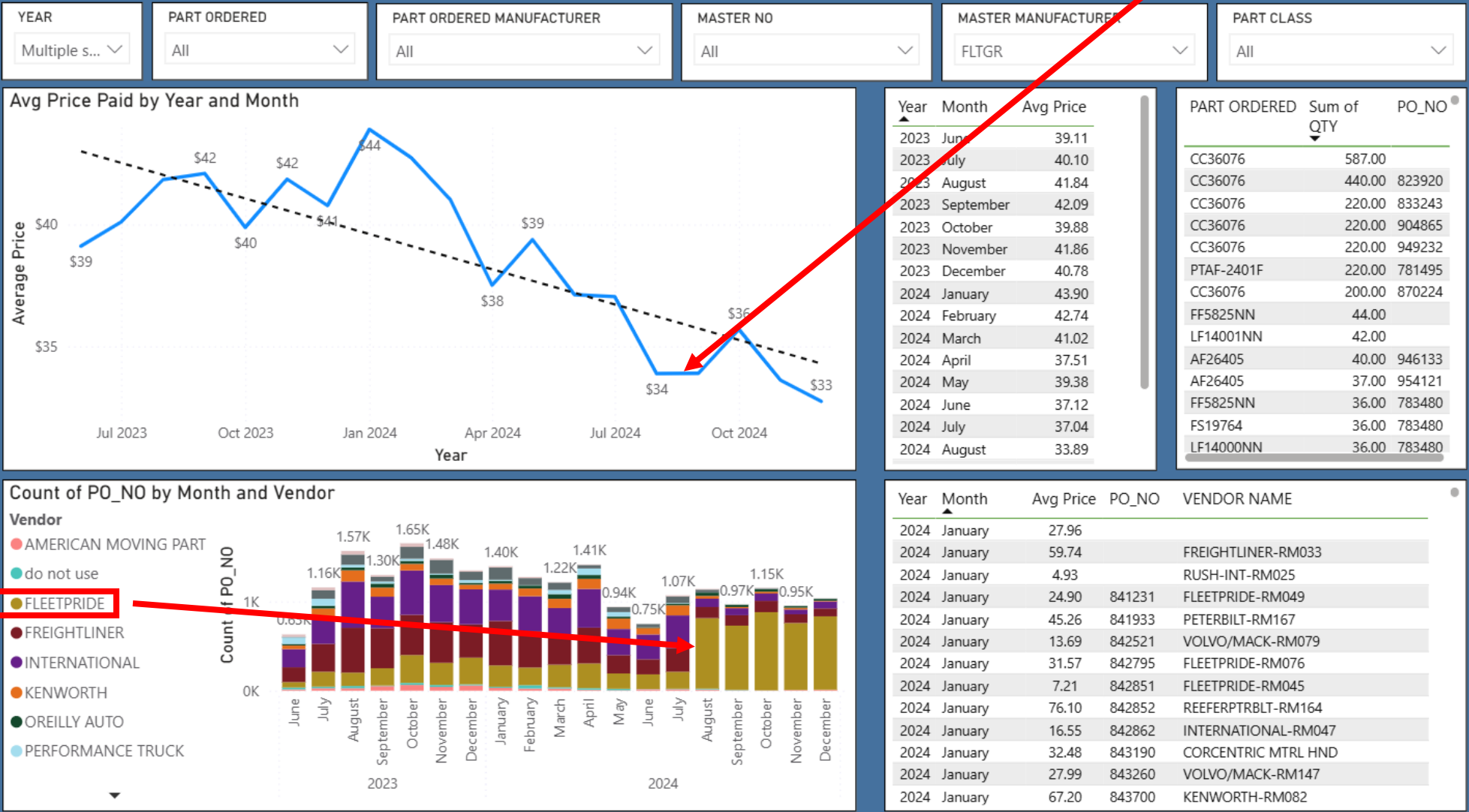
Data Inputs

- VMRS Handbook: ~900 pages
 - Mix of Tabular and Non-Tabular
- Dataset: 20,348 parts with 3rd party validated VMRS codes
 - PART, MANUFACTURER, DESCRIPTION, NOTES, MANF_PARTMFR, MANF_PARTMFR_NAME, MANF_PARTNUMBER, Match, MANF_SPARTNUMBER, VMRS
 - PART, DESCRIPTION, MANF_PARTMFR_NAME, VMRS
- Pricing details in next phase of project

Fleetguard Filters Standardization – Manufacturer & Purchasing

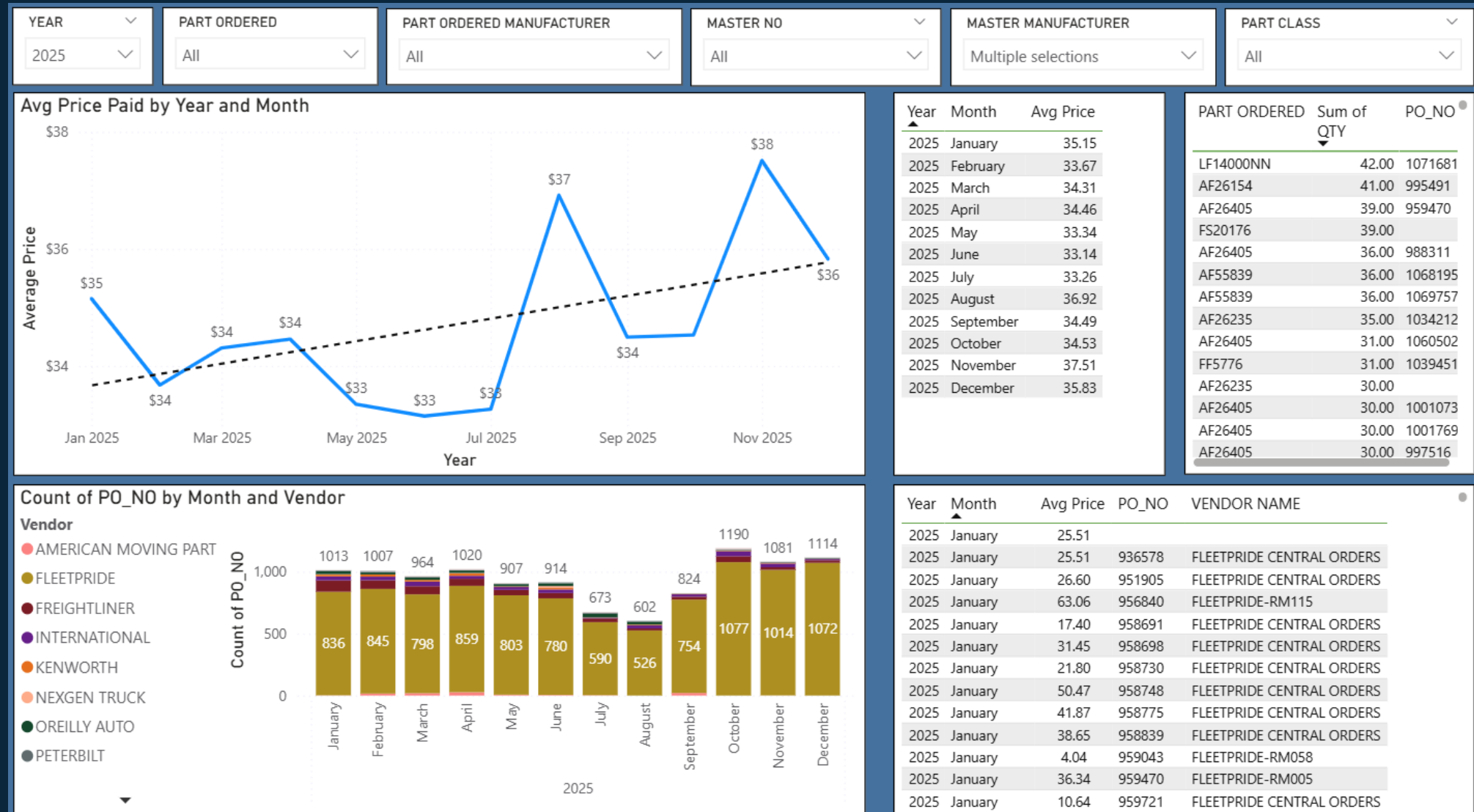
- Filters – Fleetguard Manufacturer
- Purchase via preferred supplier – Fleetpride
- Year(s): 2023-2024

Program Start Date 8/2024



Increased standard orders via Fleetpride

- Consistent Pricing (tariff impact beginning May/June 2024)
- Fleetpride Fleetguard Orders increase to 88% in 2025 vs 41% in 2024



Why choose a knowledge graph?

- Approach 1: Text matching (regex/similarity/clustering/other ML algorithm)
 - Same part, different descriptions -> no match
 - Different parts, similar descriptions -> wrong match
- Approach 2: Ranking from vendor sites
 - Inconsistent source of truth
 - No canonical authority
- Approach 3: Create a knowledge graph
 - Models the relationships and entities
 - Natively models the VMRS hierarchy
 - Traversal patterns match how users phrase questions
 - "What parts are in the Fuel System?"

What is a VMRS Code and the challenge that it solves?

- VMRS is a 9-digit combination of system, assembly, and component codes .
 - VMRS handbook created and updated yearly by: TMC (Technology and Maintenance Council) is a subdivision of ATA (American Trucking Association), the largest and most comprehensive national trade association for the trucking industry.
- From any given vendor, the fleet team may receive (1) no VMRS code, (2) an incorrect VMRS code, or (3) a correct VMRS code.
- Example: Lighting System, Lamp assembly, and LED - Marker Lamp component

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VMRS Knowledge Graph Pipeline

